Practical_4

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1 Software Lab I - Practical 4

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3 Roll No: 13

4 Batch:B1

5 Class: A8

5.1 1. Trip Cost

[ ]:
```
def hotel_cost(nights):
    return 140 * nights

def plane_ride_cost(city):
    if city == "Charlotte":
        return 183
    elif city == "Tampa":
        return 220
    elif city == "Pittsburgh":
        return 222
    elif city == "Los Angeles":
        return 475

def rental_car_cost(days):
    cost = 40 * days
    if days >= 7:
        cost = cost - 50
    elif days >= 3:
        cost = cost - 20
    return cost

def trip_cost(city, days):
    return hotel_cost(days) + plane_ride_cost(city) + rental_car_cost(days)

city = input("Enter city: ")
```

```
days = int(input("Enter number of days: "))

print("Hotel cost:", hotel_cost(days))
print("Plane cost:", plane_ride_cost(city))
print("Rental car cost:", rental_car_cost(days))
print("Total trip cost:", trip_cost(city, days))
```

5.2 2. Word Frequency

[ ]:
```
text = """Harnoor Singh struck a fine hundred before the Indian bowlers ran
through the opposition to complete a 154-run rout of UAE in their opening
U19 Asia Cup match on Thursday."""

with open("input.txt", "w") as file:
    file.write(text)

stop_words = ["the", "a", "an", "and", "is", "of", "to", "in", "on", "for",␣
 ↪"before"]

with open("input.txt", "r") as file:
    data = file.read()

words = data.split()
count = {}

for word in words:
    word = word.strip(".,!?;:'\"()")
    if word.lower() not in stop_words and word != "":
        if word.lower() in count:
            count[word.lower()] = count[word.lower()] + 1
        else:
            count[word.lower()] = 1

print(count)

keyword = max(count, key=count.get)
print("Keyword:", keyword)

with open("word_frequency.txt", "w") as file:
    for word in count:
        file.write(word + " : " + str(count[word]) + "\n")
    file.write("Keyword : " + keyword)

with open("word_frequency.txt", "r") as file:
    data = file.read()

data = data.lower()
```

```
print(data)
```

5.3 3. From Lines

[ ]:
```
data = """From xyz@rknec.edu Sat Jan 5 09:14:16 2008
From abc@rknec.edu Sat Jan 5 09:14:16 2008
From pqr@rknec.edu Sat Jan 5 09:14:16 2008
From def@rknec.edu Sat Jan 5 09:14:16 2008
lmn@rknec.edu Sat Jan 5 09:14:16 2008
tuv@rknec.edu Sat Jan 5 09:14:16 2008"""

with open("from_data.txt", "w") as file:
    file.write(data)

count = 0

with open("from_data.txt", "r") as file:
    for line in file:
        if line.startswith("From"):
            words = line.split()
            print(words[1])
            count = count + 1

print("Number of From lines:", count)
```

5.4 4. Food File

[ ]:
```
def create_file():
    with open("food.txt", "w") as file:
        file.write("Pizza|Flour, Cheese, Tomato|Mix ingredients and bake|5\n")
        file.write("Pasta|Pasta, Sauce, Vegetables|Boil pasta and add␣
 ↪sauce|4\n")
        file.write("Sandwich|Bread, Cheese, Vegetables|Put ingredients between␣
 ↪bread|4\n")

def retrieve_food(name):
    with open("food.txt", "r") as file:
        for line in file:
            data = line.strip().split("|")
            if data[0].lower() == name.lower():
                print("Name:", data[0])
                print("Ingredients:", data[1])
                print("Recipe:", data[2])
                print("Stars:", data[3])
                return

    print("Food item not found")
```

```
create_file()

name = input("Enter food item: ")
retrieve_food(name)
```

5.5 5. Student Class

[ ]:
```
class Student:
    def __init__(self, roll_no, name, marks1, marks2, marks3):
        self.roll_no = roll_no
        self.name = name
        self.marks1 = marks1
        self.marks2 = marks2
        self.marks3 = marks3

    def get_marks(self):
        return self.marks1, self.marks2, self.marks3

    def get_roll_no(self):
        return self.roll_no

    def total_marks(self):
        return self.marks1 + self.marks2 + self.marks3

    def display(self):
        print("Roll Number:", self.roll_no)
        print("Name:", self.name)
        print("Total Marks:", self.total_marks())

students = []

while True:
    roll_no = int(input("Enter roll number: "))
    name = input("Enter name: ")
    marks1 = int(input("Enter marks in subject 1: "))
    marks2 = int(input("Enter marks in subject 2: "))
    marks3 = int(input("Enter marks in subject 3: "))

    student = Student(roll_no, name, marks1, marks2, marks3)
    students.append(student)

    choice = input("Do you want to enter another student? (y/n): ")

    if choice.lower() == "n":
        break
```

```
print("\nStudent Details")

for student in students:
    student.display()
    print()
```